

Road Trip Planner — teacher reference

Three lessons of about 45 minutes; pairs rotate Planner/Checker after stages 3 and 6. Prior knowledge: rational arithmetic, expressions, graph coordinates and simple inequalities. Advanced Grade 5 / Saxon 8/7 context; use teacher judgment.

Canonical route and worked key

3 travelers; 7 days / 6 nights; 900 km; 12 km/L; \$1.60/L; one-off \$200; budget \$1,800. Fuel: 75 L, \$120. Classification: one-off and fuel once; vehicle, food, activities daily; hotel nightly.

$$C(d) = 200 + 120 + rd + h(d - 1) + 3fd + ad$$
$$m = r + h + 3f + a; b = 200 + 120 - h$$

Scenario	m / b	C(1), C(3), C(5), C(7)	Reserve / spent	Bound / whole days
Base 40/90/20/30	220 / 230	450, 890, 1330, 1770	\$30 / 98.33%	$\frac{157}{22} / 7$
Hotel 110	240 / 210	450, 930, 1410, 1890	−\$90 / 105%	$\frac{53}{8} / 6$
Food revision 15	225 / 210	435, 885, 1335, 1785	\$15 / 99.17%	$\frac{106}{15} / 7$

Base C(8) = \$1,990; revised C(8) = \$2,010. Hotel changes: +20 to m, −20 to b, +120 total over six nights. Every hotel category receives the \$20/night rise exactly once. Freeze and verify this event before permitting another choice revision.

Equivalent rules such as 220d + 230 and 200 + 120 + 40d + 90(d − 1) + 60d + 30d pass. A right seven-day total does not prove a full model. The intercept is an adjusted constant, not literal spending for a zero-day trip. Domain: whole days 1–14.

Transfer keys — teacher only

A: $T(d) = 160d + 240$, $T(4) = \$880$. B: $T(d) = 145d + 310$, $T(6) = \$1,180$. Show one learner at a time. Paper/oral submissions require a reviewer record and retain pending/verified/revise status.

Review guidance

Review reasoning quality, source truth, and oral/paper transfer separately from automatic arithmetic. Negative reserve and spending above 100% may be correct. Do not mark poor design as an arithmetic error. Any active bypass makes completion provisional. An independent correct recheck retires the active bypass while keeping history.

This reference contains keys; print separately from student proposals. No client-side teacher mode can securely conceal answers.

Choices, outcomes and classroom use

Category	Prices (USD)	Client fit points
vehicle	City compact: 30 Coastal wagon: 40 Touring van: 60	15 / 15 / 15
hotel	Trail cabin: 60 Lakeview lodge: 90 Grand hotel: 130 All quotes rise by \$20/night.	15 / 30 / 30
food	Picnic plan: 10 Local cafés: 15 Food explorer: 20	10 / 20 / 25
activities	Quiet days: 10 A little of both: 20 Adventure pass: 30	10 / 20 / 30

Outcome precedence

Missing/stale work or no recorded hotel comparison: Draft. Active bypass: Provisional. Current plan over budget or fit <65: Revision requested. Within budget and fit 65–84: Trip approved. Within budget and fit ≥85: Client delighted. Pending prose/research/transfer review is always separate.

Final event-price example	Cost / fit	Outcome
Standard car, standard hotel, full food, full activities	\$1,890 / 100	Revision requested
Standard car, standard hotel, balanced food, full activities	\$1,785 / 95	Client delighted
Basic car, basic hotel, simple food, full activities	\$1,430 / 70	Trip approved; reserve statement needed
Basic car, basic hotel, simple food, light activities	\$1,290 / 50	Revision requested despite correct affordable math
Basic car, standard hotel, balanced food, varied activities	\$1,645 / 85	Client delighted; a large justified reserve is allowed

Common misconceptions

Fuel follows the route, not the number of days. Food price is per person. Seven days includes six nights. Expand $h(d - 1)$ before collecting b . Points are (days, cost); budget is horizontal. Round fuel only at the subtotal and percentage to 0.01%; keep exact fractions for recurring bounds.

Saving and recovery

Use an alias. Export after each lesson and before moving devices. Import previews all records; copies are the default. Replacing or deleting requires a named confirmation and backup option. If device storage fails, keep the page open and export. A second tab offers reload or save as copy.

Calculator allowed for arithmetic; students must construct the model, inequality, graph and interpretation. Help examples use different numbers. Paper slips contain no keys. No timer, accounts, cloud synchronization, telemetry, or live map API.

Desktop offline and emulated-device verification do not replace a physical Safari iPad and school-network check.